

CHARGER TESTING JIG UI & FUNCTIONAL CHANGE REQUEST

CAN Monitor & Test System — Dashboard

1. Purpose

This document defines the requested UI and functional changes for the Charger Testing JIG CAN Monitor & Test System dashboard. The objective is to add temperature monitoring, correct derived-value calculations, provide tolerance-based calibration controls, introduce a controlled comparison/finalization workflow, and reduce the visual footprint of the Recent CAN Messages panel.

2. Reference Screenshot — Highlighted Areas

The following annotated image identifies the dashboard portions referenced in this change request. The red highlighted regions and numbered labels map to the requirements below.

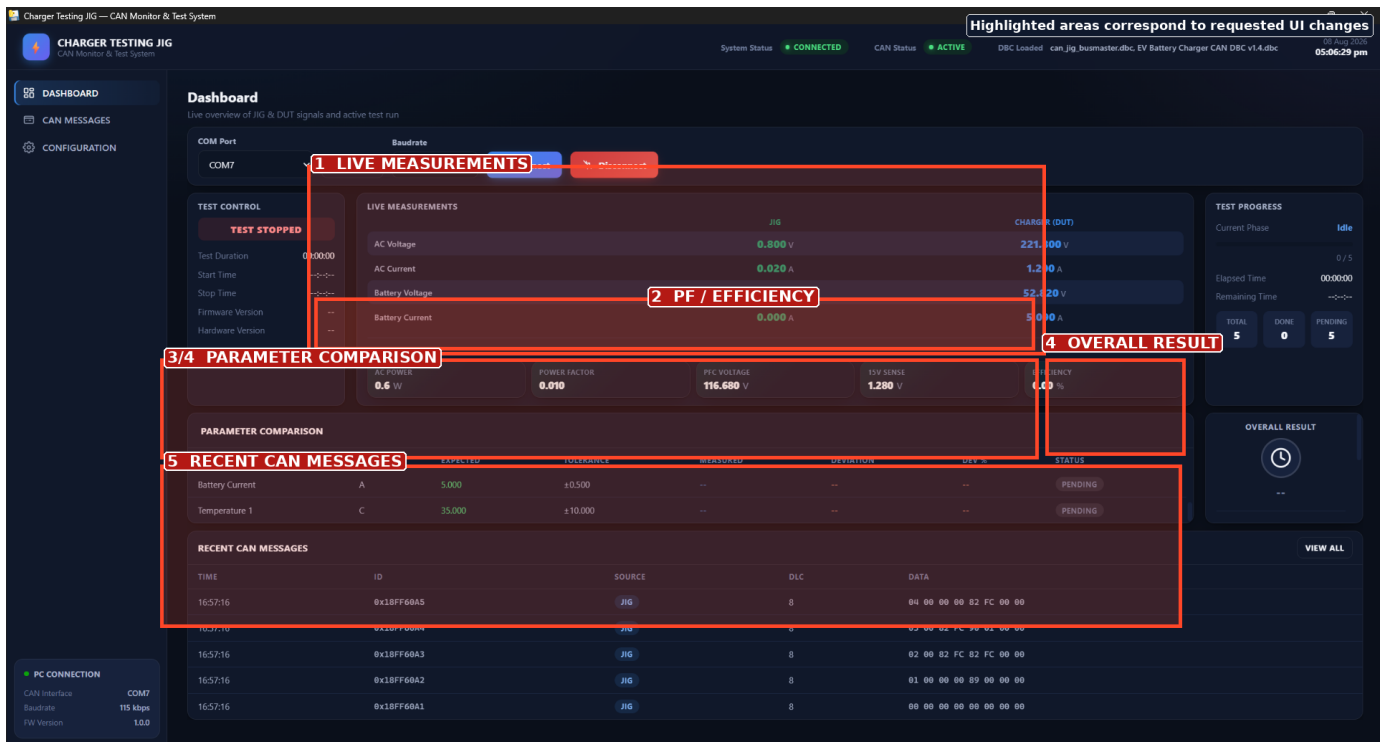


Figure 1 — Dashboard areas affected by the requested changes

3. Change Requirements

ID	Area	Requested Change	Expected Behaviour
CR-01	Live Measurements	Add temperature for both JIG and CHARGER (DUT).	Dashboard shall show temperature values under both JIG and CHARGER (DUT), using the specified CAN/DBC signals.
CR-02	Power Factor / Efficiency	Correct the division/scaling used for PF and efficiency.	Displayed values shall be calculated from the correct physical quantities and

			DBC scaling, without an incorrect fixed divisor.
CR-03	Parameter Comparison	Add calibration buttons for Battery Voltage and Battery Current.	Calibration button is visible only when the corresponding measured value is within its configured tolerance.
CR-04	Comparison / Overall Result	Add Compare and Lock controls.	Compare starts continuous parameter comparison; Lock finalizes the test and displays PASS/FAIL based on individual parameter results.
CR-05	Report	Open a report page after finalization with blank identification fields.	Report page shall contain empty fields for Charger Part Number and all required QR entries.
CR-06	Recent CAN Messages	Reduce the size/height of the Recent CAN Messages panel.	The panel shall occupy less dashboard space while retaining recent message visibility and the View All function.

4. Detailed Functional Requirements

4.1 Temperature Monitoring

Temperature shall be added to the Live Measurements section for both data sources:

- CHARGER (DUT): obtain temperature values from the ACDCParameters_1 message/signals: Temperature2, Temperature3 and TemperatureIn.
- JIG: obtain temperature from ADC_BROADCAST_TEMP1_TEMP2, using the temp1_c signal only.
- The UI should identify the temperature source clearly and display the value with °C.
- DBC signal scaling, offset, signedness and validity handling shall be applied before displaying the value.
- If a signal is unavailable or invalid, the UI should show a defined invalid/placeholder state rather than a stale value.

Recommended display arrangement

Add a Temperature row below Battery Current in the Live Measurements table, with one value under JIG and the applicable DUT temperature values under CHARGER (DUT). If three DUT temperatures are required simultaneously, use separate rows or a compact multi-value field: Temperature2, Temperature3, TemperatureIn.

4.2 Power Factor and Efficiency Calculation

The existing division/scaling for Power Factor and Efficiency is incorrect and must be corrected. The implementation should not use an arbitrary fixed divisor merely to obtain the displayed value.

Note: The exact DBC scaling factors should be taken from the project's authoritative DBC definitions. The software change should make the divisor/factor explicit rather than embedding unexplained numeric division.

4.3 Tolerance-Based Calibration Buttons

Add a calibration button beside Battery Voltage and Battery Current in the Live parameters box.

- Battery Voltage: show the Calibration button only when the measured battery voltage is within the configured tolerance of the expected value.
- Battery Current: show the Calibration button only when the measured battery current is within the configured tolerance of the expected value.
- When the measured value is outside tolerance, the corresponding Calibration button must not be visible.
- The visibility condition must update continuously while the test is active.
- Calibration action should use the currently measured value and the applicable calibration procedure; it must not be available when the input is outside the permitted range.
- The UI should prevent a hidden/ineligible calibration control from being triggered through stale UI state.

Tolerance condition

For a parameter with Expected = E, Measured = M and tolerance = T, the calibration button is eligible when $|M - E| \leq T$. For percentage tolerance, use the project's defined percentage-to-absolute conversion consistently.

4.4 Compare Button and Continuous Comparison

Add a button labelled “COMPARE” in the Parameter Comparison box.

- Initial state: comparison is inactive and individual parameter statuses remain PENDING/undefined according to the existing UI state model.
- On pressing COMPARE: enable the comparison process.
- While comparison is active, continuously evaluate each configured parameter against its expected value and tolerance.
- Update each parameter's Measured, Deviation, Dev %, and Status fields continuously as new measurements arrive.
- Individual parameter status should be at least PASS / FAIL / PENDING, with PENDING used when a valid measurement is not yet available.
- The Compare operation must not permanently finalize the overall test result; finalization is controlled separately by the LOCK button.

4.5 Lock Button and Final PASS/FAIL

Add a button labelled “LOCK” in the Overall Result box.

- LOCK shall finalize the current comparison result.
- When LOCK is pressed, evaluate all individual parameter statuses.
- Overall result = PASS only when every required individual parameter has a valid PASS status.
- Overall result = FAIL if any required parameter is FAIL.
- If a required parameter is still PENDING or invalid at the time of locking, the finalization logic should treat the test as FAIL or block finalization according to the project's selected test policy; this behaviour should be fixed and documented in software.
- After locking, the result should no longer change automatically from incoming CAN data.
- The UI should clearly indicate that the result is locked/finalized.

4.6 Report Page

After the test is finalized, open the test report page. The report page should initially contain unfilled input boxes for production/test identification data.

- Charger Part Number — blank input field.
- All required QR Code entries — blank input fields, one field per required QR.
- Test result — populated from the finalized PASS/FAIL result.

- Individual parameter results — populated from the locked comparison state.
- Test date/time and relevant test metadata — populated automatically where available.
- The report should not silently reuse old Charger Part Number or QR values unless an explicit configuration requires this.

4.7 Recent CAN Messages Panel

The Recent CAN Messages section currently occupies a large portion of the lower dashboard. Reduce its height so that it becomes a compact monitoring panel.

- Keep the latest CAN messages visible in a compact table.
- Retain Time, ID, Source, DLC and Data columns.
- Retain the VIEW ALL button for opening the full CAN message list.
- Prefer a fixed compact row count (for example, the latest 3–5 messages) rather than a large scrollable area on the dashboard.
- The reduced panel must not overlap or reduce usability of the Parameter Comparison and Overall Result sections.

4.8 Password protection on configuration

- The expected and tolerance value configuration change should be protected under password.
- A default configuration values should be there and if someone change it then it will be store somewhere.

5. Recommended Test Workflow

Step	Action	System Behaviour
1	Connect	Connect to the JIG CAN interface and confirm CAN status is ACTIVE.
2	Acquire	Receive JIG and DUT measurements, including the newly added temperature signals.
3	Compare	User presses COMPARE; individual parameters begin continuous PASS/FAIL evaluation.
4	Calibrate	Battery Voltage/Current calibration buttons appear only while the measured value is within tolerance.
5	Lock	User presses LOCK to freeze the individual statuses and calculate the final overall PASS/FAIL.
6	Report	Open the report page with blank Charger Part Number and QR entry fields for operator input.

6. Acceptance Criteria

1. Temperature2, Temperature3 and TemperatureIn are received from ACDParameters_1 and displayed under DUT; JIG temp1_c from ADC_BROADCAST_TEMP1_TEMP2 is displayed under JIG.
2. PF and efficiency match independently verified calculations using calibrated physical values and the authoritative DBC scaling.
3. Battery Voltage and Battery Current calibration buttons are visible only when their respective measurements satisfy the configured tolerance condition.
4. COMPARE enables continuous comparison and continuously updates individual parameter statuses.
5. LOCK freezes the test result and produces a deterministic overall PASS/FAIL based on the required individual statuses.

6. The report page opens after finalization and contains blank Charger Part Number and QR entry fields.
7. Recent CAN Messages is visually smaller while retaining essential recent-message information and View All access.
8. No incoming CAN update changes a test after it has been locked.

7. Implementation Notes / Open Points

- Confirm the exact DBC scaling/offset for every temperature, power, voltage and current signal before implementation.
- Confirm whether all three DUT temperatures should be shown simultaneously or whether one is the primary display value.
- Confirm the exact number and names of QR fields required on the report page.
- Define whether LOCK is allowed when any required parameter is PENDING. Recommended production-test behaviour is to prevent a false PASS.
- Define whether calibration changes only a software offset/gain or triggers a separate MCU calibration command.

8. Change Summary

The requested changes are intended to make the dashboard suitable for controlled charger production/testing: measurements become more complete, derived values become physically correct, calibration is restricted to valid measurement conditions, comparison and finalization become explicit operator actions, and the final test result is transferred into a structured report workflow.